

Phishing Awareness

Defending the Human Perimeter in 2026

CodeAlpha Cyber Security Internship | Task 2

The Threat Landscape

Why Phishing remains the #1 initial access vector.

| The Evolution of Phishing

AI-Powered Attacks

GenAI is used to create flawless, localized phishing content that bypasses traditional spam filters and linguistic detection.

Multi-Channel Strategy

Attackers combine Email, SMS (Smishing), and Voice (Vishing) to create coherent and highly believable threat scenarios.

| Modern Phishing Vectors



Quishing

QR codes in fake posters that lead to credential harvesting sites.



Vishing & Deepfakes

AI voice cloning mimicking executives to authorize fund transfers.



Smishing

Fraudulent texts claiming delivery failures or bank security alerts.

| Social Engineering Tactics

- ⚡ **Psychological Pressure:** Using artificial urgency or threats of account suspension.
- 👤 **Trust Exploitation:** Mimicking high-authority brands like Microsoft or Amazon.
- 🧠 **Pretexting:** Crafting a believable story to manipulate victims into sharing data.



| The Cost of Deception

90%

of breaches involve human error

Phishing is the Primary Entry Point

Phishing remains the primary gateway for ransomware and data breaches. In 2024, the average Business Email Compromise (BEC) loss reached \$150,000 per incident.

| Spotting the Red Flags

Element	What to look for	Risk Level
Sender Address	Lookalike domains (e.g. netflix.com)	High
Hyperlinks	Mismatch between displayed text and URL	Critical
Greetings	Generic "Dear Customer" salutations	Medium
Attachments	Double extensions (e.g. invoice.pdf.exe)	Critical

| The Deepfake Disaster

Arup Case Study (2024): A deepfake video call mimicking a high-level executive led to a £20 million unauthorized fund transfer.

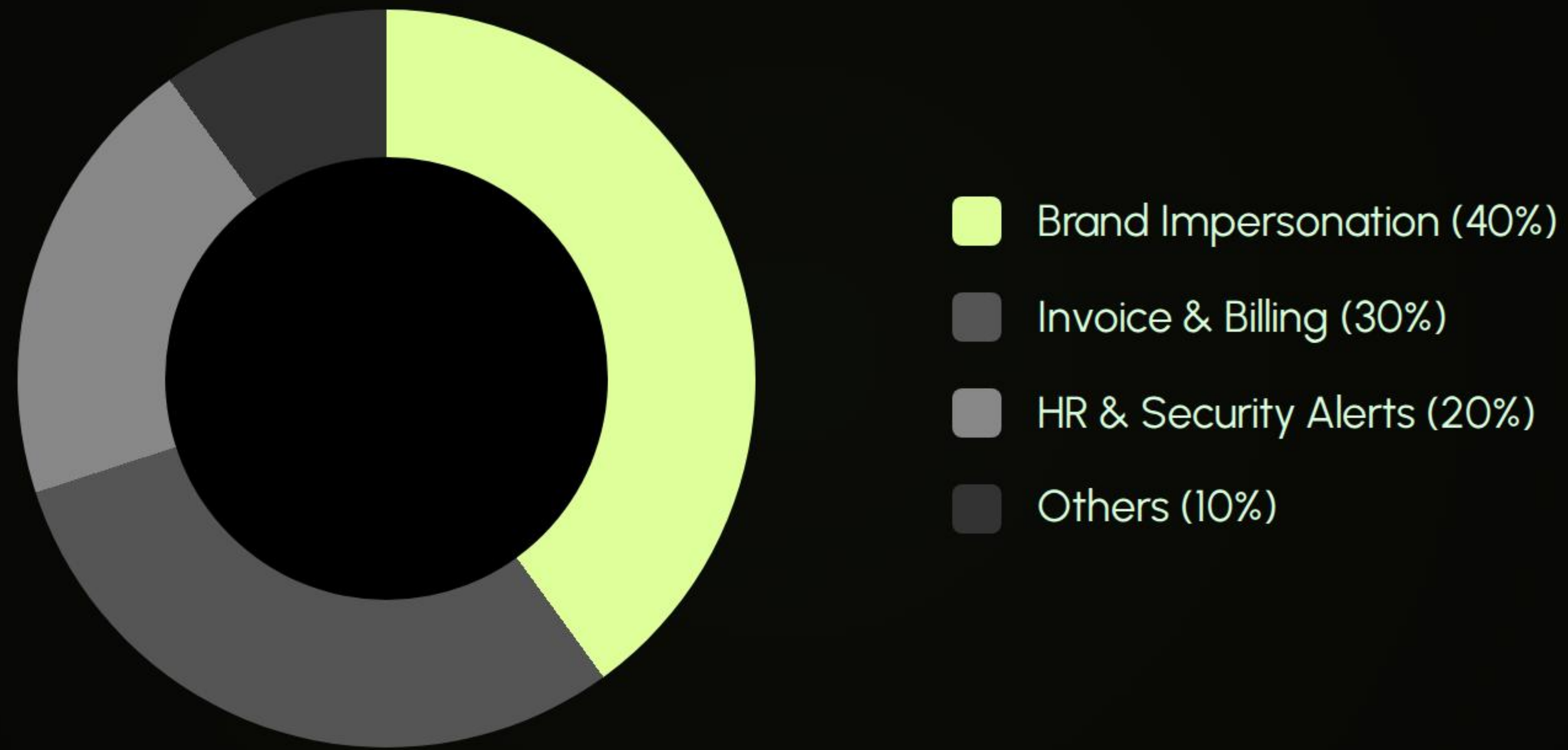
Lesson: Verification processes must evolve beyond visual and audio recognition in the age of AI. Multi-factor approval is mandatory.



| Defense **Best Practices**

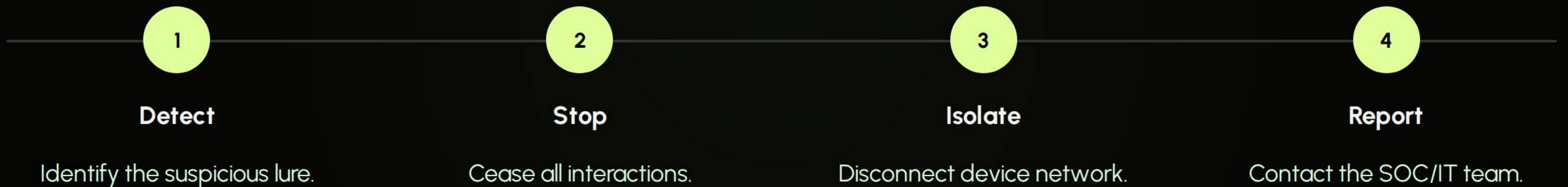
-  **Enable MFA:** Use app-based or hardware MFA everywhere possible.
-  **Hover Before Clicking:** Always inspect the destination URL before interacting.
-  **Verify Unusual Requests:** Call the sender on a known number for high-risk requests.
-  **Strong Identity:** Use password managers to eliminate credential reuse.

| Phishing Themes Proportions



Attackers leverage trust in high-authority brands to bypass suspicion.

| Incident Response Workflow



Stay Safe Online

Thank you for your attention.

CodeAlpha Cybersecurity Awareness Program
www.codealpha.tech

| Image Sources



<https://www.morphisec.com/wp-content/uploads/2019/03/Hacker-in-Mask.jpg>

Source: www.morphisec.com



<https://ironscales.com/hubfs/Blog/2025/zoom-ceo-eric-yuan-avatar-deepfake.webp>

Source: ironscales.com